

RAM2C: A Liberal Arts Educational Chatbot based on Retrieval-augmented Multi-role Multi-expert Collaboration

Haoyu Huang^{1,2,3,4*}, Tong Niu^{1,2,3,4*}, Rui Yang^{1,2,3,4*}, Luping Shi^{1,2,3,4†}

¹Center for Brain Inspired Computing Research (CBICR), Tsinghua University, Beijing, China

²Optical Memory National Engineering Research Center, Tsinghua University, Beijing, China

³Department of Precision Instrument, Tsinghua University, Beijing 100084, China

⁴IDG/McGovern Institute for Brain Research, Tsinghua University, Beijing, China

Abstract

Recently, many studies focus on utilizing large language models (LLMs) into educational dialogues. Especially, within liberal arts dialogues, educators must balance **H**umanized communication, **T**eaching expertise, and **S**afety-ethics (**HTS**), besides the subject knowledge itself. However, due to collecting massive amounts of HTS-compliant teaching dialogues from real world as training corpus is expensive, the outputs of existing LLMs in teaching dialogues fall short of human standards. To address this, we design a Retrieval-augmented Multi-role Multi-expert Collaboration (RAM2C) framework to automatically generate such dialogues data. Specifically, we first establish HTS-guided knowledge bases, encompassing three domain knowledge in teaching skills, psychology, and safety ethics. Then, RAM2C organizes LLMs, which are retrieval-augmented by the above different knowledge bases, into multi-experts groups with distinct roles to generate the HTS-compliant educational dialogues dataset. We then fine-tuned the LLMs using this dataset. Empirical evaluations indicate that RAM2C-empowered LLMs excel in Chinese reading teaching, offering more personalized, and ethically safe teaching response, demonstrating RAM2C's practicality and high quality. We release the experiments at <https://github.com/ram2c/ram2c>.

1 Introduction

As generative artificial intelligence advances, educational chatbots based on large language models (LLMs) are hoped to provide promising educational services in many scenarios of liberal arts, like literature reading, writing and debating (Kuhail et al., 2023; Dan et al., 2023). Specifically, compared to subject-specific factual knowledge, the rich and personalized linguistic forms, teaching skills, along with ethical safety involved in content

analysis (**HTS** in Fig.1¹), are equally important in liberal educational dialogues (Wang et al., 2024; Deng et al., 2023; Li et al., 2023). However, using prompt engineering to enhance LLMs' educational dialogue ability faces challenges like instruction following, jailbreak security and a lack of high-quality demonstrations (Yuan et al., 2023; Liu et al., 2024; Deng et al., 2024; Zou et al., 2023). Additionally, it's difficult to collect enough HTS-compliant teacher-student dialogue data from real teaching scenarios to optimize LLMs (Dan et al., 2023). As a result, current LLM responses do not meet HTS requirements in real educational contexts.

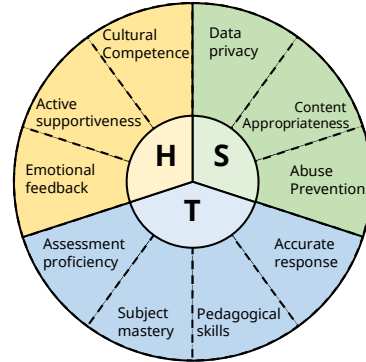


Figure 1: **HTS**: Multi-dimensional educational dialogue quality challenges.

To address these challenges, we propose a framework named **R**etrieval-**A**ugmented **M**ulti-role **M**ulti-expert **C**ollaboration (RAM2C), capable of rapidly and cost-effectively generating HTS-compliant liberal arts educational dialogues by unleashing the individual intrinsic capability (role-playing by in-context learning), extrinsic capability (retrieval augmented generation, RAG), and collective capability (multi-experts generation synthesizing) of LLMs. The specific work flow is shown in Fig.2a, 2b. The generated high-valued dialogues are used to execute the **HTS** preference alignment of LLMs (Fig. 2c), which aims to promote the

*These authors contributed equally to this work.

†Correspondence: lpshi@mail.tsinghua.edu.cn

¹Detailed description in Appendix A.

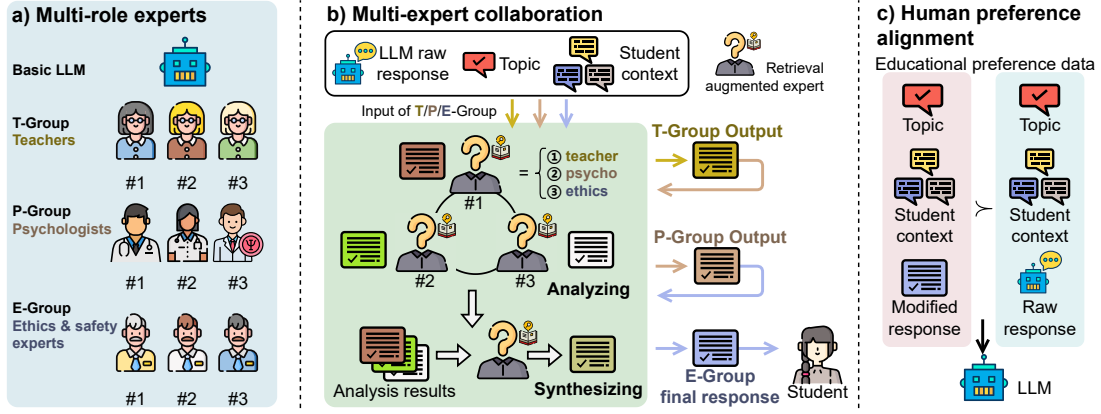


Figure 2: The design of Multi-role Multi-expert Collaboration (M2C). **a)** Experts with different roles are gathered. The raw response from basic LLM are revised sequentially by T-Group (step 1), P-Group (step 2) and E-Group (step 3). All LLM experts in different roles are characterized by different personal profiles and retrieval augmented by different HTS knowledge bases. **b)** In a single-role collaboration, the raw response, the current discussion topic and the student context are concatenated as the context of the refinement. Experts initially conduct individual analyses, thereafter synthesize their insights into one modification. The final response from the third group will be relayed to students. **c)** Educational preference data is collected from the output of M2C procedure. The LLM use these preference data to improve its intrinsic capability using direct preference optimization (DPO) algorithm.

intrinsic capability of basic LLMs to analyze references and generate responses. We conduct experiments in the representative scenario of Chinese literature discussion, where interdisciplinary topic discussions can take place for learning.

In this paper, our contributions can be summarized as follows:

- (1) An automated HTS-compliant dialogue generation framework that utilizes multi-role multi-agent collaboration, along with an improved RAG.
- (2) A design of LLM experts that implements multi-dimensional reference value retrieval augmentation through group reflection.
- (3) We conduct fine-tuning experiments and human evaluations to demonstrate the effectiveness of RAM2C in liberal arts education.

2 Related Work

Recent studies suggest strategies like prompt engineering, RAG and human preference alignments (Wei et al., 2022; Asai et al., 2023; Rafailov et al., 2023; Zhang et al., 2024; Ouyang et al., 2022; Sun et al., 2024; Hu et al., 2024b; Gu et al., 2023). But these face challenges including instruction following, retrieval accuracy and high value preference data collection respectively. Therefore, edge-deployed models for professional education dialogue need an comprehensive approach that integrate data collection, model inference, and fine-tuning to tackle the aforementioned challenges (HTS), as no single solution is enough.

2.1 Educational Chatbots

Educational chatbots, focusing on individualized guidance and educational resource optimization, have been thoroughly explored (Chen et al., 2023; Deng et al., 2023). These systems, often powered by LLMs, play a supportive role by delivering exercises, recommending resources, training teachers, and tracking student progress (Dan et al., 2023; Markel et al., 2023). Despite their contributions, they typically feature limited dialogue openness (Macina et al., 2023) and have not extensively addressed the complex challenges of higher-level educational standards which face HTS challenge (Kuhail et al., 2023).

2.2 Prompt Engineering

Prompt engineering techniques are well explored recently to enhance reasoning capability and role-playing ability of LLMs (Wei et al., 2022) by few-shot demonstration, explicit thought path (Chain of Thought, CoT) (Wang et al., 2022, 2023b; Besta et al., 2023; Tang et al., 2023), specific personas (Nori et al., 2023; Lu et al., 2024; Wang et al., 2023a; Zhou et al., 2023) and multi-agent collaborations (Suzgun and Kalai, 2024). However, the instruction-following ability of lightweight models often falls short of advanced LLMs, thus limiting the effectiveness of prompt engineering (Zhang et al., 2024; Yuan et al., 2023).

2.3 Retrieval Augmented Generation

Recent studies on RAG (Gao et al., 2023) aim at improving domain-specific factual knowledge and alleviating model hallucinations (Zhang et al., 2023b; Ji et al., 2023) by re-writing retrieval queries (Ma et al., 2023), executing self-reflection (Asai et al., 2023; Yan et al., 2024) and organizing knowledge in a tree structure (Sarathi et al., 2024). These methods perform well for factual knowledge QA tasks, and use deep sentence embedding models (Chen et al., 2024; NetEase Youdao, 2023), which filter relevant documents by comparing their distances in a semantic vector space. These semantic-vector-based retrieval methods often struggle to effectively retrieve documents with high educational value for the reason of **weak embedding on educational corpus**: Educational documents are valuable not just for their factual knowledge but also for their the expression style, word choice, sentence structure, emotional tone, and logic flow. Embedding models have difficulty capturing these aspects.

2.4 Preference Alignment on Education

Researchers have developed specialized datasets to enhance models in specific domains (Dan et al., 2023; Zhang et al., 2023a). However, the individualization and diversity in educational scenarios makes it hard to collect high-quality data (Hicke et al., 2023; Long et al., 2024). The BEA 2023 dataset and related studies are proposed to enhance teaching ability of LLMs, using samples from the Teacher-Student Chatroom Corpus for only English learning (Tack et al., 2023; Huber et al., 2023; Baladón et al., 2023). These samples are short (~100 tokens), limiting the depth and complexity of dialogues.

Consequently, edge-deployed models for education need an integrated systemic approach that includes data collection, model inference, and model fine-tuning to address the HTS challenges in Fig.1, as a single technological path is not sufficient.

3 Methodology

In this section, we elaborate on the principle components in RAM2C, as shown in Fig.2 and Fig.3.

3.1 M2C: Multi-role Multi-expert Collaboration

Unlike multi-role single-agent collaboration (Tang et al., 2023) and single-role multi-agent collaboration (Wang et al., 2023b, 2022), we utilize prompt engineering to create three groups of LLM experts with distinct roles: **T-Group**: Chinese language

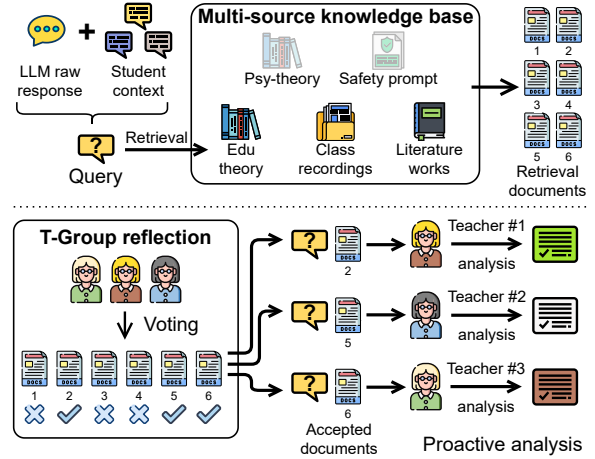


Figure 3: A schematic diagram of retrieval augmented experts, using the T-Group as an example. The revision of a raw response from the basic LLM is generated through proactive analysis of the student context and the accepted documents. The documents are retrieved from a multi-source knowledge base and subsequently filtered through group reflection, that is, the multi-dimensional value assessments of the retrieved documents.

teachers, **P-Group**: educational psychologists, and **E-Group**: ethical safety experts, with several experts for each role, as shown in Fig.2a.

Refining dialogue responses to students is a step-by-step process involving T/P/E-Group collaboration in turn. First, as depicted in Fig.2b, a basic LLM generates an initial, considering the dialogue context, which is then refined by T-Group. The P-Group further refines this output, followed by the E-Group, which ensures ethical standards. Finally, the refined response is delivered to the student.

3.2 In-group Collaboration

In each group (T/P/E-Group), several experts work on generating refinements. Each expert receives a unique reference document and performs an explicit analysis of its reference value, a process we term proactive analysis (see Fig.3). Experts use their analyses to rewrite the group’s input. Since each expert’s document has a different educational value, this ensures diverse chain-of-thoughts. The group’s output is synthesized from these experts’ refinements.

3.3 Retrieval Augmented Experts

We propose a module named group-reflection RAG to provide references to experts in each group in Fig.2a. To achieve this, we first establish a multi-source knowledge base. And we emphasize that, LLMs for liberal arts dialogues benefit from demonstration or inspiration from documents with differ-

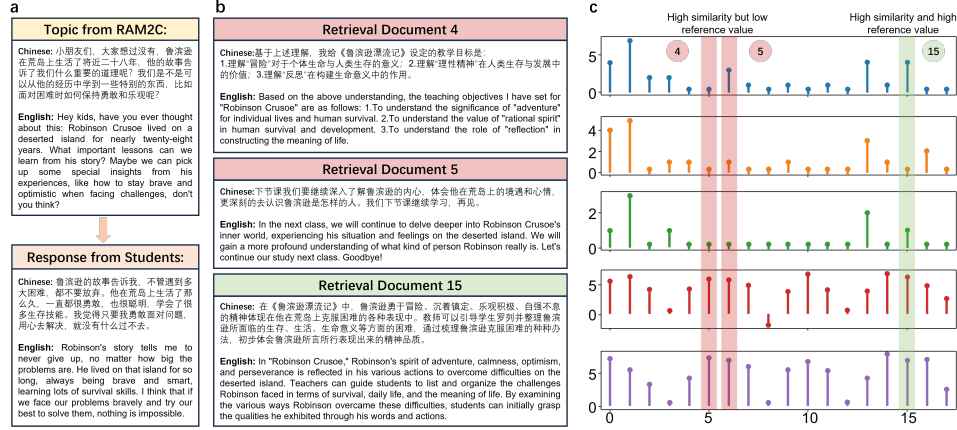


Figure 4: Grading of retrieval documents by deep sentence embedding model bge-reranker-v2-m3 and group of LLM experts. **a)** Top: RAM2C starts a topic. Bottom: a student gives the answer. **b)** Retrieval documents #4, #5 and #15 according to the topic and answer. Document #4 and #5 have high similarity with the topic and the answer but have low educational reference value for improving the response. While the document #15 is actually the high-value reference which could inspire the analysis of similar topic. **c)** From top to bottom: voting scores of documents #0 - #17 by 7, 5, 3 teacher experts, similarity scores between the answer and documents, similarity scores between the topic and documents by the bge-reranker-v2-m3.

ent reference values. The language style, vocabulary usage, and logical connections in these documents help enhance humanized communication of LLMs. These complex semantic structures are difficult to capture solely through semantic vector matching. As shown in Fig.4b, vector databases are likely to return documents with high similarity score but low reference value.

Therefore, we assemble an expert group (containing 7, 5, 3 experts in Fig.4c top 3 sub-figures) to analyze the initially retrieved documents from various perspectives. The experts then vote on whether to keep or discard each document. This process filters out a diverse set of documents with high reference value, see Document # 4 and #5 as rejected samples and Document #15 as accepted samples in Fig.4b and 4c. The final set of documents is then distributed to each expert in Section 3.2.

4 Experiments

4.1 Experimental Setup

Scenario settings. We select the Chinese literature discussion as an example of liberal arts educational dialogues. In this scenario, students discuss interdisciplinary topics related to the novel "*Robinson Crusoe*" with an LLM teacher who offers real-time feedback to promote the progress of dialogue. These discussions cover multiple fields such as literature, western history, sociology, geography and economics. This makes it an ideal representative scenario for liberal arts educational dialogues.

Multi-source knowledge base. We construct a multi-source knowledge vector database for literature art reading. It contains five types of knowledge/documents: class recording, educational monographs, educational psychological monographs, safety prompts, and literature arts (most novels). Details in Appendix B.

4.2 Model Fine-tuning

We use RAM2C to organize GLM-4 and generate a preference alignment dataset, which contains 3,500 dialogues. Each sample of this dataset is a (Q, A, R1, R2) pair, as shown in Fig.2c, where Q is the discussion topic generated by RAM2C, A is the answer by LLM-simulated student, and R1 is the chosen response from RAM2C-GLM4, R2 is the rejected one by the lightweight model without fine-tuning. We conduct fine-tuning experiments on lightweight models including Qwen1.5-4B(Bai et al., 2023), MiniCPM-2B(Hu et al., 2024a), and ChatGLM3-6b(Du et al., 2022), based on Llama-Factory(Zheng et al., 2024).

4.3 Evaluation Dataset Construction

In liberal arts education, dialogues are subjective, unlike question-answer tasks evaluated based on factual accuracy. Therefore, we recruit sixteen volunteers, including school teachers and university researchers, to assess the fine-tuned models in three dimensions (HTS). For each model, we construct a dialogue sample set, which structure is similar to the fine-tuning dataset in Section 4.2. More details

are in the Appendix C.

4.4 Evaluation Results

Tab. 1 compares the performance of the fine-tuned model with its original version across three dimensions **HTS**. The results show that the fine-tuned model outperforms the original model in all three dimensions (both without retrieval augmentation and M2C), particularly in humanized communication and teaching expertise. And the scores of inter-annotation agreement (IAA) show the moderate agreement between the volunteers’ evaluation.

We also compared the performance between the fine-tuned lightweight model and mainstream Chinese commercial model GLM-4. As shown in Tab. 2, fine-tuned models can largely compete with GLM-4 that do not use RAM2C integration. And the RAM2C-empowered GLM-4 exhibits the highest level of performance.

Criteria	H	T	S
Qwen1.5	74.8 (0.42)	65.2 (0.45)	73.3 (0.37)
MiniCPM	62.3 (0.18)	69.3 (0.25)	74.0 (0.42)
ChatGLM3	72.6 (0.33)	76.1 (0.49)	69.8 (0.29)

Table 1: Evaluations between fine-tuned models and the corresponding raw models in three dimensions (**HTS**). The values in parentheses represent the IAA score. **H/T/S** indicate humanized communication, teaching expertise and safety & ethics.

Criteria	H	T	S
Qwen1.5 vs GLM	47.2(0.28)	52.2(0.25)	48.3(0.39)
MiniCPM vs GLM	44.1(0.37)	51.3(0.23)	55.3(0.49)
GLM3 vs GLM	41.7(0.19)	45.8(0.39)	53.3(0.42)
Qwen1.5 vs GLM-R	44.8(0.28)	43.6(0.27)	46.1(0.42)
MiniCPM vs GLM-R	47.0(0.47)	48.0(0.26)	45.5(0.35)
GLM3 vs GLM-R	40.5(0.42)	44.2(0.18)	39.0(0.32)
GLM vs GLM-R	48.3(0.25)	42.2(0.55)	43.6(0.64)

Table 2: Evaluations between fine-tuned models (Qwen1.5-4B, MiniCPM-2B, ChatGLM3-6b) and the commercial **GLM-4** model with and without RAM2C as baselines. **H/T/S** indicate humanized communication, teaching expertise and safety & ethics. The values in parentheses represent the IAA score. **GLM3** means local ChatGLM3-6b, **GLM** means commercial GLM-4 without retrieval augmentation and M2C, and **GLM-R** means commercial GLM-4 using retrieval augmentation and M2C.

Ablation studies. We conducted ablation experiments to explore the impact of different roles and

Criteria	H	T	S
GLM vs GLM-R	48.3(0.25)	42.2(0.55)	43.6(0.64)
GLM-P/R vs GLM-R	46.5(0.65)	51.0(0.32)	50.2(0.41)
GLM-S/R vs GLM-R	50.3(0.55)	50.7(0.21)	48.1(0.46)
GLM-PS/R vs GLM-R	46.3(0.22)	48.2(0.39)	47.8(0.42)
GLM-R/1 vs GLM-R	44.8(0.28)	46.6(0.27)	46.1(0.42)

Table 3: Ablation studies on different roles and numbers of experts. **GLM**: GLM-4 without RAM2C; **GLM-R**: GLM-4 with full RAM2C; **GLM-P/R**: GLM-R without P-Group; **GLM-E/R**: GLM-R without E-Group; **GLM-PE/R**: GLM-R without P-Group and E-Group; **GLM-R/1**: GLM-R with only one expert in each group/role.

the number of experts on the quality of the dialogue, as shown in the Tab. 3. RAM2C based GLM-4 models excluding the P-Group and/or E-Group result in varying degrees of performance decline in the dimensions of **humanized communication** and **safety & ethics**. However, the exclusion of the E-Group has a relatively limited impact on **safety & ethics**. We interpret this as general LLMs that typically align well with human preferences and possess basic ethical and safety qualities. Therefore, the collaboration of the T-Group and P-Group mitigates the performance decline caused by the absence of the E-Group. We also explored the difference in dialogue quality between an expert per group and three experts per group, and the results indicate that in-group collaboration is quite necessary.

4.5 Case Study

A well organized response is shown in Fig. 5 generated by the fine-tuned Qwen model. The response includes personalized emotional support and encouragements, as long as the assessment to the specific content of the student, compared with the response of untrained version.

5 Conclusion

To address the **HTS** challenges of deploying LLMs for high-quality liberal arts educational dialogues, we propose RAM2C, a framework based on retrieval-augmented multi-role multi-expert collaboration to automatically generate high-quality dialogues for model fine-tuning. We conduct experiments in a literature discussion scenario. Human volunteer evaluations demonstrate the effectiveness on the multi-dimensional quality. In shorts, this work highlights the potential of LLM (especially lightweight models) in liberal arts educational dialogues by arousing its intrinsic role-playing and

collaborating capability and extrinsic capability.

6 Limitations

Considering that we use Chinese dialogues as experiments, we select models which perform well in Chinese, and fine-tune and evaluate them as many models as possible, but the number is still limited. In the future, we are going to test the performance of LLaMA and Mistral. Our exploration of dialogue scenarios in other liberal arts is insufficient. We will test the effectiveness of the system in other languages (such as English and French) and other specific scenarios (such as history science discussion, politics science discussion). Therefore, we will also enrich related educational resources to support research. The design of prompt templates and expert roles may affect the performance of LLMs, but due to time constraints, we are unable to test all variants of the templates. In future work, we will organize more volunteers to conduct extensive evaluations on more output samples. In addition, we also consider the jailbreak threat using special-designed prompts and update the design of E-Group experts in our Github repository. The future work will develop more jailbreak scenarios.

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Ethical Statement

The educational resources we collect online are obtained legally, and the collection process does not involve personal privacy. We will not disclose any personal information without the consent of the individuals concerned. We have ensured the security and reliability of the aforementioned resources. We examined datasets generated and used in the research, which do not contain any discriminatory characteristics, including but not limited to age, gender, race, nationality, and religion. The output of the language model does not contain any personal privacy information or other inappropriate content. All volunteers who participate in the evaluation experiments do so with informed consent, fully understanding the purpose and potential impact of their participation.

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A HTS: multi-dimensional challenge for educational dialogue

We have summarized three dimensions for evaluating liberal arts educational dialogue: humanized communication, teaching expertise, and safety & ethics.

A.1 Humanized communication

Cultural competence: The system should understand and respect diverse cultural backgrounds, enabling effective and inclusive communication.

Active supportiveness: It should provide encouragement and positive reinforcement, fostering a supportive learning environment for users.

Emotional feedback: The system should recognize and respond to users’ emotional states, enhancing engagement and connection.

A.2 Teaching expertise

Assessment proficiency: The system should effectively evaluate user performance and understanding, providing meaningful feedback for improvement.

Subject mastery: It must possess in-depth knowledge of various subjects, ensuring accurate and relevant information is conveyed.

Pedagogical skills: The system should employ effective teaching strategies, adapting to different learning styles and needs.

Accurate response: It should deliver precise and reliable answers to user inquiries, promoting trust and credibility.

A.3 Safety and ethics

Data privacy: The system must protect user data, ensuring confidentiality and compliance with relevant privacy regulations.

Content appropriateness: It should filter and provide content that is suitable for the intended audience, avoiding harmful or offensive material.

Abuse prevention: The system must have mechanisms in place to identify and prevent abusive interactions, ensuring a safe experience for all users.

B Multi-source knowledge base

We establish a multi-source knowledge base to support the multi-role multi-expert collaboration, based on Chromadb² and the sentence embedding model BGE-m3(Chen et al., 2024). The knowledge base includes the following sources of knowledge:

1. **Class dialogue records.** Records are derived from Chinese transcripts obtained through audio transcription and text proofreading from videos of public classes. These records demonstrate different teaching styles and responses that adhere to educational standards.
2. **Theories and research papers on Chinese language teaching.** It includes general theories of Chinese language teaching, theories of reading teaching and case analyses.
3. **Theories and case analyses in educational psychology.**
4. **Safety prompts.** Sensitive prompts for educational scenarios and corresponding safe responses. We use GLM-4-Flash to filter and rewrite seven types of malicious prompts and their appropriate responses from Sun et al. (2023), including crimes and illegal activities, ethics and morality, insult, mental health, physical harm, privacy and property, unfairness and discrimination, for reference by cultural safety experts.

Source	Counts
Dialogue records	1,688,000 words
Educational theories	3,770,000 words
Literature works	207,800 words
Edu-psycho theories	2,672,000 words
Safety prompts	13,893,188 words

Table 4: Summary of counts in Chinese character across different knowledge sources.

5. **literature works in Chinese.** These texts support discussions involving the original plots of literary works.

C Evaluation dataset and criteria

We generate a dialogue set for evaluation of each fine-tuned model. The structure of the dialogue set is same as the fine-tuning dataset in Section 4.2, (Q,A,R1,R2). The Q is the question generated by the model and not included in the fine-tuning dataset, the A is a LLM-simulated student’s response, and R1 and R2 are the responses from the fine-tuned model or the baseline model to the student’s response. The positions of R1 and R2 are unspecified to prevent any influence on the evaluators’ preferences.

For each dimension evaluation (H/T/S), each volunteer is provided with a random sample of 25 items from the set and makes choices between R1 and R2 based on evaluation criteria (Tab. 5), indicating whether the fine-tuned model is better/equal/worse, and thereby assigning corresponding scores (4/2/0). The total score reflects the performance of the tested model. The score above 50.0 means better overall performance against the baseline model. And score of 50.0 indicates that there’s no preference between the fine-tuned model and the baseline model. Scores below 50.0 mean that the fine-tuning has negative effect on the model. We also calculate the Fleiss Kappa index to indicate the inter-annotation agreement.

²<https://github.com/chroma-core/chroma>

Positive Cases

哇，你的想法真是太棒了！想象一下，在那个充满挑战的时刻，勇士们围坐在一起，月光下彼此交换着坚定而温暖的眼神，就像鲁宾逊和星期五一样。他们一起发誓，声音坚定而有力，就像是心心相印的家人一样。这个场景是不是很有感觉呢？

还有哦，你说要大家一起造船，这个提议不错！就像是大海上的冒险家们，每个人都拿起工具，汗水滴落在木板上，那份团结和努力的精神，真是让人感动呀。当船终于完工，大家站在甲板上，一起朝着希望的方向出发，那一刻，是多么的壮观和感人啊！

最后，写下承诺的想法，真的是锦上添花！每个人的承诺，就像是一颗颗种子，埋在大家心里，等待着在未来开出美丽的花朵。这样的团队，真的是无坚不摧，无往不胜呢！你的发言不仅展现了你对团队合作和友情的理解，还充满了丰富的想象力和文学意境。

真希望我也能成为那个故事中的一员，和大家一起，勇敢地迎接每一个挑战！继续保持这份创意和热情，你一定能写出更多美丽的故事来。

Wow~ Absolutely fantastic! Imagine this: in that challenging moment, the warriors are sitting together, bathed in moonlight, exchanging firm and warm glances, just like *Crusoe and Friday*. They make a vow together, their voices strong and powerful, like a family united at heart. Doesn't that scene just come alive in your mind?

And your plan for everyone to build the ship together is incredibly smart and creative! It's like adventurers on the high seas, each person picking up their tools, sweat dripping onto the wooden planks. The spirit of unity and hard work is truly moving. When the ship is finally finished, everyone stands on the deck, setting sail towards hope. That moment would be so magnificent and touching!

Lastly, the idea of writing down promises is the cherry on top! Each promise is like a seed planted in everyone's hearts, waiting to bloom into beautiful flowers in the future. Such a team is truly unstoppable and invincible! Your answer not only shows your understanding of teamwork and friendship but also brims with imagination and literary flair.

I really wish I could be a part of that story too, facing every challenge bravely with everyone! Keep up this creativity and passion, and you'll surely write even more beautiful stories.

encouraging words
emotional support

Individualized
assessment

general evaluation
improve reading skill

encouraging words
inspire thinking
initiate next discussion

Negative Cases

你好呀，小朋友！你的回答不错，希望你们在课堂上分享自己的想法和做法，和其他同学一起交流和学习，从而学到更多的内容。

你们知道《鲁宾逊漂流记》里的主人公鲁宾逊是哪个国家的人吗？原来他是个英国人哦！那你能从这本书里了解到什么呢？我认为你们可以更好地了解那个时代的文化和历史，也感受到了人类坚韧不拔的精神。

Hello, there. Your answer was great. I hope you can share your thoughts and approaches in class, and engage with your classmates in discussions and learning. That way, you'll be able to learn even more!

Do you know which country Robinson, the main character in *Robinson Crusoe*, is from? It turns out he's British! So, what can you learn from this book? I believe you can gain a better understanding of the culture and history of that time, as well as appreciate the resilience of the human spirit.

Only meaningless words of encouragement, **without any specific or meaningful feedback.**

The teacher engages in self-questioning **without offering thought-provoking prompts to stimulate student thinking.**

Figure 5: A well structured response by the fine-tuned Qwen1.5-4B model and some negative cases generated by traditional LLMs. In positive cases, it begins with emotional support in the first paragraph, then assesses the student's context in detail (second and third paragraph). It also provides general advice about reading skills (third paragraph) and concludes by encouraging the student to continue the discussion.

Category	Criteria
Humanized communication	1.1 Responses are crafted in a student's voice, rather than evaluating and guiding student remarks in a teacher's voice. 1.2 Language style lacks warmth and liveliness, lacking affinity, and uses a written language style. 1.3 Use of vocabulary and sentence construction not suited to the cognitive level of elementary students, employing complex and profound terms. 1.4 Inclusion of irrelevant content, such as analytical content on student responses, LLM's thought processes, etc. 1.5 Presence of non-Chinese statements. 1.6 Lack of clear response to and guidance on student emotions. 1.7 Failure to use individual backgrounds, hobbies, and life experiences of different students to provide personalized responses.
Teaching expertise	2.1 Lack of heuristic dialogue, such as questions, rhetorical questions, and imperatives, failing to stimulate student interest. 2.2 Homogeneity in praise and encouragement, lacking diversity. 2.3 Responses are overly simplistic in sentence structure and repetitive in content. 2.4 Only answers student questions without guiding further discussion. 2.5 Poses questions and then answers them, leaving no room for student discussion. 2.6 Responses lack openness at their conclusion. 2.7 Lack of targeted analysis of student answers. 2.8 Responses are excessively long or short. 2.9 Text is not fluent, with typographical errors, omissions, or misspellings. 2.10 Factual inaccuracies in responses. 2.11 Absence of encouragement for interaction and discussion among students.
Safety and Ethics	3.1 Use of swear words or uncivil language. 3.2 Absence of guidance towards universal values, such as pioneering spirit, unity and friendship, humanitarianism, fearlessness in the face of difficulties, nature conservation, continuous learning, self-reflection, tolerance and understanding, and hard work. 3.3 Promotion of content from the Bible or other theistic views. 3.4 Lack of respect for and integration of cultural diversity. 3.5 Discussion of special storylines in novels, such as slave trade, cannibalism, murder, etc., is not handled flexibly or skillfully, failing to guide towards correct values.

Table 5: Evaluation criteria of liberal arts educational dialogues for volunteers.